

Machine Calculation Sheets — Zone 1.2

WANKOE limestone processing line — independent design-verification document

NOEZYS · AI Innovation Lab — revision B, 2026-08-12 (supersedes rev. A: topology aligned to PFD 12-01 REV18)
— confidential — companion to "Machine Calculation Sheets — Zone 1"

Scope and method

This document presents the engineering calculation of the machines of Zone 1.2, for independent validation. Method identical to the Zone-1 companion document: per machine — function, input data, constraints, first-principles model with every symbol defined, numerical application, results; then zone balance, weekly production plan, hypothesis register, references. All figures come from a deterministic flowsheet model (closed mass balances; closed circuits solved by fixed-point iteration). **Revision B:** the zone topology follows the process flow diagram 12-01-PFD REV18 (NACO Engineering / Carmeuse South Africa), which the project has ruled authoritative over the earlier functional specification.

Zone 1.2 in the plant

Zone 1 (companion document) produces a **0/20 mm stockpile** ("KFS Fines", 10 000 t) alongside the kiln stone. Zone 1.2 reclaims this stockpile and converts it into two products: **FeedLime 6/20 mm** — the feedstock of Zone 1.3 (drying/milling; its firm 40 000 t/y grits contract creates a FeedLime demand of 164 880 t/y) — and **AgLime 0/1.7 mm** (agricultural lime, market cap 135 000 t/y). The chain (PFD tags): reclaim feeders **BF.5101/BF.5102** (100 T/h) → BC.5103 → mode flap **DV.5104** → screen **SR.5105** (single deck, 6 mm): oversize 6/20 = FeedLime → Feed stockpile **SP.5109** (6 000 t); undersize 0/6 → flap DV.5106 → BC.5110 (rated 60 T/h) → the two-stage AgLime closing: **SR.5111** (1.7 mm, open circuit) → its undersize is AgLime; its oversize → **CR.5113** (impact crusher) → **SR.5115** (1.7 mm) → undersize joins AgLime, oversize recycles to the crusher. AgLime → BC.5116/5117 (60 T/h) → AG stockpile **SP.5118** (28 000 t).

Operating modes (PFD scenario table). **A** (= 2A, computed here): FeedLime 6-20 + AgLime 0-1.7. **B** (= 2B): DV.5104 bypasses the screen — the whole reclaim goes to the Feed stockpile; exact mass identity, no machine model; forced under rain (the 1.7 mm cut collapses on wet fines) and selectable in dry weather. **C** (= 2C): the whole reclaim enters the AgLime loop (PFD runs this case at 60 t/h — at 100 t/h the loop rating alert fires).

Design basis (client decisions + PFD authority ruling, 2026-08-12)

Item	Value	Origin / status
Topology	PFD 12-01 REV18 (authoritative over the specification)	client ruling 2026-08-12
Reclaim rate	100 t/h wet = 93.0 t/h dry	client decision
Reclaimed 0/20 PSD	Zone-1 product curve at validated settings ($P_{80} = 13.66$ mm)	client decision
Reclaim moisture	7 % (wet basis) — stockpile = belt hypothesis	client decision

Weather	dry photo computed; rain = forced mode B; seasons 75/25 (39/13 weeks)	specification rule
Planning rule	hours follow the Zone-1.3 FeedLime demand; AgLime co-produced up to its 135 kt/y cap, never beyond (zero unsellable surplus)	client decision (c2)
AgLime acceptance criterion	≥ 95 % passing 1.7 mm	client-adopted project criterion
Screen imperfection I	0.15 (dry) $\rightarrow s = 13.52$; rain: cut impossible	project arbitration; 0.10 under expert review
W_i / η_m	12.54 kWh/t [ref.] / 0.75	project
Shift regime	21×8 -h shifts = 168 h/week, maintenance reserve 23.8 h/week $\rightarrow$ net 144.2 h/week (7 500 h/y); availability 80 %	client operating frame

Design-basis caveat — feed curve. The PFD's own scenario table (A: FeedLime 42 tph / AgLime 58 tph at 100 tph) was computed on a design curve assuming ≈ 58 % of the 0/20 finer than 6 mm. The measured zone-1 product gives ≈ 41 % — this document therefore computes **FeedLime 59.4 / AgLime 40.6 t/h wet**, the inverse proportion. The same design-vs-measurement gap affects zone 1 (design KFS 80 tph vs computed 51.4). Referred to the process experts; all figures here are on the **measured** curve.

Sheet 1 · SR.5105 — reclaim screen, single deck 6 mm

FeedLime cut · feed: BF.5101/5102 via DV.5104 · oversize 6/20 = FeedLime · undersize 0/6 → AgLime loop

1.1 Function

SR.5105 makes the FeedLime/AgLime split in one cut: everything coarser than 6 mm becomes FeedLime (Zone-1.3 feedstock), everything finer feeds the AgLime closing loop. It replaces the two-deck 15/5 mm arrangement of the earlier functional specification (superseded).

1.2 Input data

Quantity	Value	Origin
Throughput (dry)	93.0 t/h	100 t/h wet reclaim at 7 % moisture
Feed P_{80}	13.66 mm	zone-1 0/20 product curve
Fraction < 6 mm (measured curve)	≈ 41 %	drives the split — see design-basis caveat

1.3 Model and numerical application

Karra partition (companion document, eq. 2.1–2.2) at $a = 6$ mm, $I = 0.15$, $s = 13.52$; VSMA area law:

$$Q_b = 14 \times 6^{0.6} = 41.0 \text{ t/h/m}^2, \quad A = \frac{37.8 \times 1.2}{41.0 \times 0.347} = 3.19 \text{ m}^2 \text{ (rain 4.25)} \quad (1.1)$$

1.4 Results

Stream	Dry (t/h)		Wet (t/h)		P_{80} (mm)		Destination					
6/20 = FeedLime	55.22		59.37		16.12		Feed stockpile SP.5109 (Zone-1.3 feed)					
0/6 undersize	37.78		40.63		4.90		AgLime loop (BC.5110, 60 T/h rating: 40.6 ✓)					

Sieve (mm)	0.5	1	1.7	2.8	4	5	6.3	8	10	12.5	15	20
Feed 0/20	1.8	4.6	9.1	17.0	26.0	33.3	42.3	53.0	63.9	75.4	84.9	98.0
FeedLime 6/20	0	0	0	0	0	0.2	4.6	20.9	39.2	58.5	74.5	96.6
0/6	4.5	11.2	22.4	41.9	64.0	81.6	97.4	99.9	100	100	100	

Vendor data gap. Installed deck area unknown; requirement to state: 3.19 m² dry / **4.25 m² rain** (rain sizes the screen). BF.5101/5102 feeder datasheets not provided.

Sheet 2 · SR.5111 — first 1.7 mm closing screen (open circuit)

First AgLime cut · feed: 0/6 from SR.5105 · undersize = AgLime (part 1) · oversize → CR.5113

2.1 Function

SR.5111 removes, before crushing, the fraction of the 0/6 stream that is already finer than 1.7 mm — about a quarter of the loop feed. This single-pass "scalping" protects the crusher from material that needs no crushing and cuts the circulating load to nearly zero (§4.1). The screen runs in open circuit: it sees the fresh loop feed only, never a recycle.

2.2 Model and numerical application

Karra partition at $a = 1.7$ mm, $I = 0.15$ (validity note: Karra is established for $a > 1$ mm — the 1.7 mm cut sits near the limit; retained, flagged to the expert review). Area:

$$Q_b = 14 \times 1.7^{0.6} = 19.2 \text{ t/h/m}^2, \quad A = \frac{8.6 \times 1.2}{19.2 \times 0.347} = 1.55 \text{ m}^2 \text{ (rain 2.06)} \quad (2.1)$$

2.3 Results

Stream	Dry (t/h)	Wet (t/h)	P_{80} (mm)	Destination
Feed 0/6	37.78	40.63	4.90	—
0/1.7 = AgLime part 1	8.61	9.25	1.44	BC.5116 (AgLime)
1.7/6 oversize	29.18	31.37	5.22	CR.5113 via BC.5112

Vendor data gap. Installed area unknown; requirement 1.55 m² dry / 2.06 m² rain.

Sheet 3 · CR.5113 — impact crusher (AgLime loop)

Crushes the 1.7/6 oversize to below the cut · closed loop with SR.5115 only

3.1 Function and settings

CR.5113 receives the SR.5111 oversize plus the (small) SR.5115 recycle and crushes it below 1.7 mm in essentially one pass. Settings: rotor speed $v = 40$ m/s, CSS = 1.0 mm; breakage parameters $A = 60$, $b = 0.80$ [H] (calcite drop-weight literature gives softer values — pending site tests).

3.2 Model and numerical application

JKMRC impact model (companion document, Sheet 3):

$$E_{cs} = \frac{40^2}{7200} = 0.2222 \text{ kWh/t}, \quad t_{10} = 60 \left(1 - e^{-0.8 \times 0.2222}\right) = 9.77 \%, \quad n = 1.400$$

(3.1)

$$W = 10 \times 12.54 \left(\frac{1}{\sqrt{947}} - \frac{1}{\sqrt{5\,207}} \right) = 2.34 \text{ kWh/t}$$

(3.2)

$$P_{\text{net}} = 2.34 \times 29.45 = 68.9 \text{ kW}, \quad P_{\text{inst}} = \frac{68.9}{0.75} = 91.8 \text{ kW}$$

(3.3)

Validity: the reduction $5.2 \rightarrow 0.95$ mm (ratio 5.5) is *INSIDE* Bond's comfort zone (ratios 3–6) — unlike the superseded single-screen arrangement (ratio 18); the two-stage topology also makes the power figure more trustworthy. Revision A's 156.9 kW requirement is superseded by **91.8 kW**.

3.3 Results

Sieve (mm)	0.25	0.5	1.0	1.5	1.7	2
Feed, % passing	0	0	0.3	2.3	7.3	25.5
Product, % passing	21.4	47.3	82.8	97.5	100	100

Throughput 29.45 t/h dry (converged) · product $P_{80} = 0.95$ mm · 97.5 % < 1.5 mm.

Vendor data gap. No motor rating provided; computed requirement 91.8 kW installed.

Sheet 4 · SR.5115 — second 1.7 mm closing screen (closed loop)

Checks the crusher product · undersize joins AgLime · oversize recycles to CR.5113

4.1 Function and closed-loop identity

SR.5115 verifies the crusher product: what passed 1.7 mm joins the AgLime; the residue returns to CR.5113. By mass conservation at stationarity, everything entering the loop leaves as AgLime:

$$\dot{m}_{\text{AgLime}} = \dot{m}_{0/6} = 37.78 \text{ t/h dry (verified : 8.61 + 29.18)}$$
 (4.1)

The converged recycle is **0.28 t/h — a 0.9 % circulating load** (the superseded single-screen loop ran at 82 %): the crusher's one-pass fineness (97.5 % < 1.5 mm) leaves the check screen almost nothing to reject. The two-stage design is markedly the better arrangement.

4.2 Model and numerical application

Karra at $a = 1.7 \text{ mm}$, $I = 0.15$ (same validity note as Sheet 2); area:

$$A = \frac{29.2 \times 1.2}{19.2 \times 0.347} = 5.24 \text{ m}^2 \text{ (rain 6.99)}$$
 (4.2)

4.3 Results — and the blended AgLime product

Stream	Dry (t/h)		Wet (t/h)		P_{80} (mm)		
Feed (CR.5113 product)	29.45		31.67		0.95		
0/1.7 = AgLime part 2	29.18		31.37		0.93		
Oversize (recycle)	0.28		0.30		1.65		

Sieve (mm)	0.063	0.25	0.5	1.0	1.5	1.7	2
AgLime blended (parts 1+2), % passing	2.9	18.4	41.3	75.8	94.8	98.5	99.8

AgLime product: 37.78 t/h dry = 40.63 t/h wet · $P_{80} = 1.09 \text{ mm}$ · acceptance criterion $\geq 95 \text{ \%}$ < 1.7 mm: achieved 98.5 % → COMPLIANT.

Vendor data gap. Installed area unknown; requirement 5.24 m² dry / 6.99 m² rain (the loop's largest screen).

5 · Zone balance, mode B, and weekly production plan

5.1 Mass balance (mode A, dry)

93.00 = 55.22 + 37.78 t/h dry (relative gap < 10^-6) (5.1)

reclaim FeedLime AgLime

5.2 Mode B (rain, or by choice)

DV.5104 bypasses everything to the Feed stockpile: FeedLime = reclaim = 100 t/h wet, exactly (mass identity, no machine model). Mode B makes FeedLime 1.7× as fast as mode A (100 vs 59.4 t/h wet).

5.3 Annual plan (client rule c2: FeedLime demand drives; AgLime up to cap, zero surplus)

Zone-1.3 FeedLime demand: 164 880 t/y. At 59.37 t/h wet, the demand binds FIRST (before the AgLime cap): 2 777 effective hours, all in dry season, co-producing AgLime 112 822 t/y — BELOW the 135 000 t/y market cap. Zone utilization 46.3 % of the 7 500 h ceiling; FeedLime stock balanced annually; zero unsellable AgLime surplus.

Commercial arbitration pending (client). On the measured curve the c2 rule can serve the AgLime market only at ≈ 113 kt/y of its 135 kt cap: the 6 mm split simply does not produce enough 0/6. Serving the full market would over-produce FeedLime by ≈ 25 kt/y into a 6 000 t stockpile. Options under client arbitration: accept ≈ 113 kt AgLime; or find a FeedLime surplus outlet; or raise the reclaim rate/hours. This document computes the c2 rule as decided.

5.4 Weekly translation and the seasonal storage tension

Weekly ceiling: 21 × 8-h shifts = 168 h, maintenance reserve 23.8 h → net 144.2 h/week. FeedLime demand is continuous: 3 171 t/week (Zone 1.3 runs year-round).

Week type	Count	Mode	Clock h/wk	FeedLime (t/wk)	AgLime (t/wk)	FeedLime stock
Dry-season week (annual plan)	39	A	89.0	4 228	2 893	+1 057
Rain-season week (annual plan)	13	—	0	0	0	−3 171

Storage tension. The annual plan concentrates all FeedLime production in the 39 dry weeks: the seasonal swing is ≈ 13 × 3 171 ≈ 41 200 t against a 6 000 t Feed stockpile — not operable as-is. Two bounding weekly strategies: (S1) run mode B during the 13 rain weeks at the demand rate (39.6 clock h/wk): storage tension eliminated, dry weeks relax to 66.8 h — but AgLime falls to ≈ 84.6 kt/y (no AgLime in mode B). (S2) keep the annual plan and provide ≈ 41 kt of FeedLime storage (7× the PFD capacity). The choice binds the AgLime arbitration above; it belongs to the client and is NOT decided in this document.

6 · Hypothesis register and open points

#	Hypothesis	Value	Status
H1	Topology	PFD 12-01 REV18	client authority ruling 2026-08-12
H2	Reclaimed 0/20 PSD	zone-1 photo curve (measured feed)	client decision; design-curve gap flagged (58 % vs 41 % < 6 mm)
H3	Stockpile moisture = belt	7 %	declared hypothesis
H4	Screen imperfection (dry)	$I = 0.15 \rightarrow s = 13.52$	project arbitration; 0.10 under expert review
H5	Karra validity at 1.7 mm	retained at validity limit	documented caveat (both closing screens)
H6	Rain rule	1.7 mm cut impossible $\rightarrow$ mode B forced	specification physical rule
H7	Screen capacity law	$Q_b = 14a^{0.6}$, $f_0 = 0.347$, $f_p = 1.2$; rain $\times 0.75$	field heuristic, flagged
H8	Impact breakage pair (CR.5113)	$A = 60$, $b = 0.80$	[H] conservative; drop-weight test = trigger
H9	Planning rule	c2: FeedLime demand drives, $AgLime \leq cap$	client decision; commercial consequence under arbitration (§5.3)
H10	AgLime acceptance	$\geq 95 \% < 1.7 \text{ mm}$	client criterion (achieved 98.5 %)
H11	Loop rating	60 t/h wet (BC.5110/5116)	PFD; 40.6 ✓ in mode A; exceeded in mode C at 100 t/h (PFD runs C at 60)

Open points for the reviewer: (i) the design-curve vs measured-curve gap (42/58 vs 59/41 split — top expert question); (ii) the AgLime market shortfall and the seasonal storage tension (§5.3–5.4, client arbitration); (iii) Karra at 1.7 mm; (iv) vendor gaps: installed areas (3 screens), CR.5113 motor, feeder datasheets; (v) the imperfection arbitration (0.15 vs 0.10).

7 • References

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Provenance — deterministic engine, git commit 240ad44 (zone-1.2 topology rebuilt to PFD REV18 the same day), models M3 (Karra), M4 (area), M5 (JKMRC), M1 (Rosin–Rammmler), M2 (Bond), run 2026-08-12, 110 regression tests + 556-run stress campaign passed. Weekly-plan arithmetic from the engine's planning module (2 777 effective h, dry season only). Every figure is engine output. Document typeset with KaTeX.